RIDHI SRIKANTH

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SUMMARY

Computer Engineering student with systems experience spanning runtimes, operating systems, and networking. Experienced building and debugging low-level software in C/C++, Python, and Shell, with interests in usable abstractions and reliable infrastructure.

EDUCATION

University of California, San Diego | BS, Computer Engineering | GPA: 3.975 **Graduation: June 2027**
BS/MS Applicant (Expected MS Graduation: June 2028)

- **Selected Courses:** Data Structures, Algorithms, Operating Systems, Networks, Parallel Computing, Computer Architecture
- **Awards and Honors:** 1st Place SPLASH SRC 2026, CSE Dept. Honors Candidate, HKN IEEE Honor Society; Provost Honors
- **Selected Programs:** Brown Systems Week; Cornell–Maryland–Max Planck Research School (CMMRS) 2026

EXPERIENCE

Research Lead | SALAD Lab UCSD | Advisor: Professor Michael Coblenz
A Python–Shell Hybrid Programming Model *June 2025 - Present*

- Extended Python with native shell constructs, enabling Python programs to directly execute shell commands and compose system-level workflows through pipelines and I/O redirection
- Extended CPython in C across parsing, AST construction, bytecode generation, and evaluation runtime loop to implement the new language features and execution model
- Engineered interoperability between Python and shell processes, including persistent shell execution and automatic conversion between Python and shell data representations
- Diagnosed and resolved compiler and runtime failures using GDB, LLDB, AddressSanitizer, and low-level execution tracing
- Validated the extended interpreter against 33K+ CPython tests, debugging runtime regressions
- Built a shell-to-extended-Python translation system to evaluate expressiveness against real world shell scripts
- Designed and conducted a controlled user study comparing the Python–Shell system with standard Python approaches to evaluate effectiveness and usability
- Extended Abstract (link) accepted and **won first place at SPLASH/ISSTA SRC 2026**; full paper under review

Research Assistant | ATLAS Lab Brown University | Advisor: Professor Nikos Vasilakis
Stream Monitor Injection *July 2026 - Present*

- Contributed to RT, a type system for shell programs, with new type rules, inference heuristics, and richer type abstractions.
- Developing runtime monitoring infrastructure to detect silent failures in shell pipelines using static and dynamic analysis
- Building a benchmark of real-world shell pipeline bugs from Stack Overflow and GitHub commits, reproducing bugs, environments, fixes, and monitored programs to evaluate detection

Full-stack Developer – ElectroTriton With Professor Curt Schurgers | Electrotriton *Aug 2025 – Present*

- Developing features in ElectroTriton, a Django and MySQL homework management web app used for Engineering courses
- Implemented course cloning and automated roster updates, reducing management time by ∼71% and ∼78%, respectively
- Developed a new analytics dashboard for instructors to track assignment metrics, including view, start, and completion rates.

Research Assistant | SALAD Lab UCSD
Scientific Workflow Reproducibility Study *Feb 2025 - June 2025*

- Collaborated with a research team on a corpus study of 500+ scientific programs
- Analyzed selected programs for runtime errors, dependency issues, and execution failures affecting software reproducibility
- Contributed to a research manuscript on scientific workflow reproducibility: link to pre-print

Course Tutor *Sep 2025 - Present*

- **CSE 110 Software Engineering**: Guiding students through TypeScript labs, emphasizing software design practices, evaluating code architectures and implementations
- **ECE 35 (Analog Design):** Guiding 400+ students on circuit theory, breadboard hardware implementations and debugging

SKILLS

Programming Languages: Python, C, C++, TypeScript/JavaScript, SQL, Shell, Java, OpenCL
Systems & Tools: Linux, GDB, LLDB, AddressSanitizer, Git, Docker, CI/CD
Technologies: React, Flask, Django, Firebase, Agile

LEADERSHIP

- **IEEE UCSD Vice Chair Events (2026–27):** Lead the events team in organizing technical, professional, outreach, and social events for the UC San Diego IEEE community, partnering with industry and campus organizations.
- **IEEE UCSD Technical Chair (2024-26):** Led 5+ technical workshops for 100+ students, including live-coding sessions on React and recommendation systems